Name: ____________________

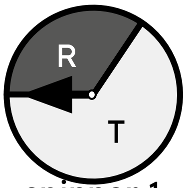
Worksheet

Calculating theoretical probabilities from probability trees (two events)

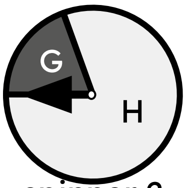
Task A: Probability of an outcome from a two-stage trial

1) Spinner 1 is spun several times.
The spinner landed on R $\frac{1}{3}$ of the time.

Spinner 2 is spun several times.
The spinner landed on G $\frac{1}{7}$ of the time.



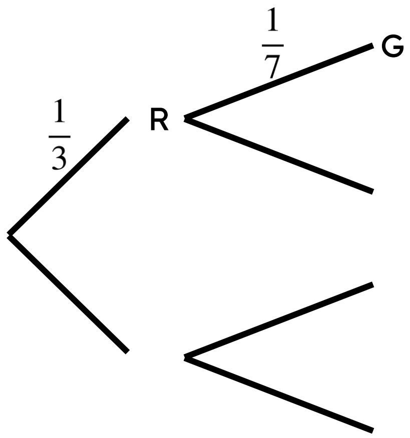
spinner 1



spinner 2

Trial: spinning spinner 1 and spinner 2 once each.

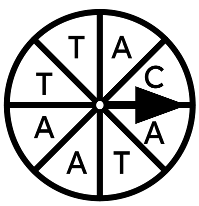
Complete the probability tree to calculate the probability of each outcome from this trial.



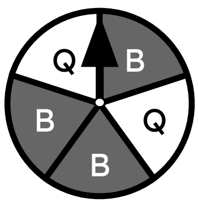
R G	$\frac{1}{3} \times \frac{1}{7} = \frac{1}{21}$
	$\frac{2}{21}$

2) Trial: spinner 1 and spinner 2 are spun once each.

This probability tree currently shows all the outcomes on spinner 1.

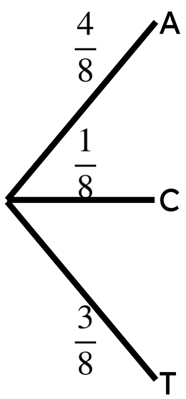


spinner 1



spinner 2

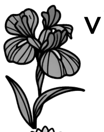
a) Modify this probability tree so that it shows the sample space and probability of each outcome of this trial happening.



b) What is the probability of the outcome C Q happening?

3) Lucas, Izzy, and Laura each take two flower seeds from a packet to grow.

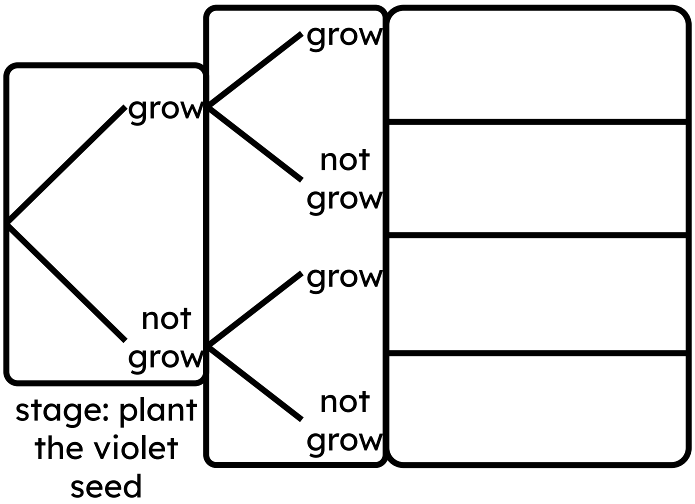
The probability of each type of flower growing is written on the packet.


violet: 80%

rose: $\frac{3}{4}$ 


daisy: 90%

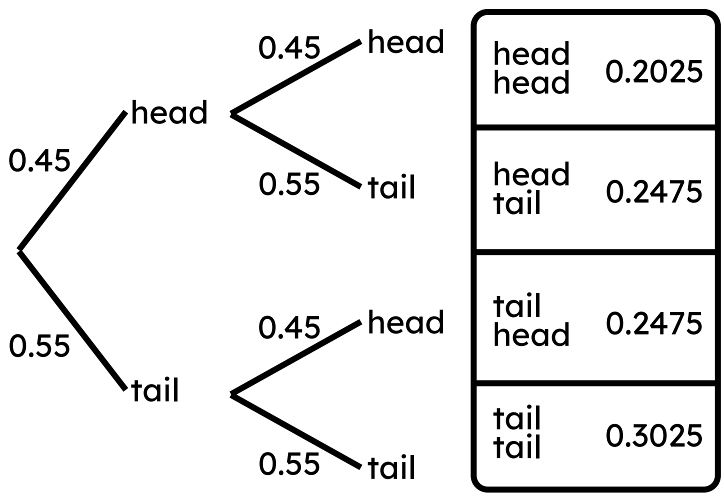
Lucas' tree



Construct three probability trees (one each for Izzy, Lucas, and Laura) which show the probabilities of their two flowers growing.

Task B: Probability of an event from a two-stage trial

1) A coin is flipped several times. On 45% (or 0.45 as a decimal) of trials, the coin lands on heads. Trial: flipping this coin twice.

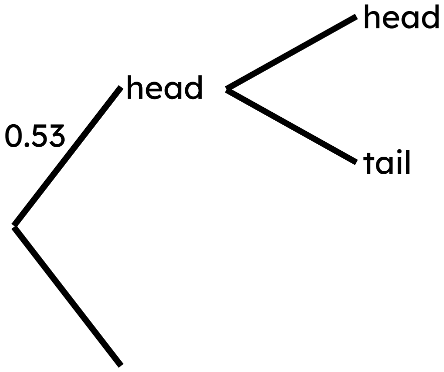


a) What is the probability of the outcome "two heads"?

b) What is the probability of the event that at least one coin lands on a heads?

c) What is the probability of the event that exactly one coin lands on a tails?

2) A different coin is flipped several times. On 53% (or 0.53 as a decimal) of trials, the coin lands on heads. Trial: flipping this coin twice.



head head
head tail
tail head
tail tail

Event 1: landing on heads twice.
Event 2: landing on exactly one head.

a) Complete the probability tree and sample space.

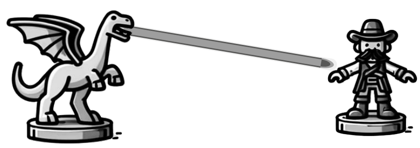
b) Which of these two events is more likely to happen?
And by how much?

3) Lucas is playing a video game, and needs to scare away a cowboy using either his dragon or unicorn.

The probability of a character hitting their opponent is shown by their “accuracy” stat.

In order for Lucas to scare away the cowboy, either Lucas’ character has to hit, or the cowboy has to miss (or both).

By constructing two probability trees, find out whether the dragon or unicorn is more likely to scare away the cowboy.



dragon accuracy: 80%

cowboy accuracy: 60%

unicorn accuracy: 60%

cowboy accuracy: 25%